

Chern-Weil forms (and abstract homotopy theory)

Week 4

Freed-Hopkins
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Previously, for a principal G -bundle $p: E \rightarrow X$
with a connection

$$CW_{p:E \rightarrow X} : \text{Sym}^\bullet(\mathfrak{g}^\vee)^\mathfrak{g} \rightarrow \Omega^\bullet(X)$$

is a homomorphism.

$$\begin{array}{ccc} \text{Let } \text{Bun}_G^\nabla : \text{Man}^{\text{op}} & \longrightarrow & \text{Grpd} \\ M & \longmapsto & \left\{ \begin{array}{l} \text{principal } G\text{-bundles} \\ \text{with connections over } M \\ \text{and connection} \\ \text{preserving isomorphisms} \end{array} \right\} \end{array}$$

$$\begin{array}{ccc} \Omega^p : \text{Man}^{\text{op}} & \longrightarrow & \text{Set} \hookrightarrow \text{Grpd} \\ M & \longmapsto & \left\{ \begin{array}{l} p\text{-forms on} \\ M \end{array} \right\} \end{array}$$

pullbacks on morphisms.

Thm: The map

$$CW : \text{Sym}^\bullet(\mathfrak{g}^\vee)^\mathfrak{g} \xrightarrow{\sim} \text{Nat}(\text{Bun}_G^\nabla(-), \Omega^\bullet(-))$$

is an isomorphism

Chern-Weil forms are the only natural differential forms associated to a connection on a principal G -bundle.

Yoneda philosophy

There exists a universal classifying 'space' Bun_G^∇ such that

$$\text{Map}(X, \text{Bun}_G^\nabla) \simeq \{\text{principal } G\text{-bundles with connection}\}$$

There exists a universal 'space' of differential forms $\Omega^\bullet$

$$\text{Map}(X, \Omega^\bullet) \simeq \{\text{deRham complex of } X\}$$

We want to interpret $\text{Map}(\text{Bun}_G^\nabla, \Omega^\bullet)$ as $\Omega^\bullet(\text{Bun}_G^\nabla)$

'Generalized smooth spaces' are $\text{Sh}(\text{Man}, \infty\text{-Grpd})$
and smooth maps

This is an ∞ -category in particular an ∞ -topos.

Ref Chapter 6
HTT.

[$\mathcal{C}$ is an ∞ -category. $\mathcal{C}$ is an ∞ -topos if there exists a small ∞ -category $\mathcal{K}$ and
an accessible left exact localization functor $\text{PSh}(\mathcal{K}) \rightarrow \mathcal{C}$]

Example to keep in mind $\text{Sh}(\mathcal{K})$ for some Grothendieck topology on $\mathcal{K}$.

Sketch :

Part 1 a) Prove that $\text{Bun}_G \simeq {}^*//G$
 b) Prove that $\text{Bun}_G^\nabla \simeq \Omega^1(-, \mathfrak{g})//G$

Part 2 for any discrete sheaf $\mathcal{F}$ with a G -action, the map
 $\Omega^\bullet(\mathcal{F}/G) \rightarrow \Omega^\bullet(\mathcal{F})$ induced by $\mathcal{F} \rightarrow \mathcal{F}/G$
 effective epimorphism

is injective and the image can be easily characterized as "special G -invariant forms on $\mathcal{F}$ ".

Part 3 $\Omega^\bullet(\Omega^1(-, \mathfrak{v})) \simeq \text{Kos}^\bullet(\mathfrak{v}^\vee)$

Part 4 $\Omega^\bullet(\Omega^1(-, \mathfrak{g})//G)$ is ' G -invariant' sub-complex of $\text{Kos}^\bullet(\mathfrak{g}^\vee)$

Show that this subcomplex is isomorphic to $\text{Sym}^\bullet(\mathfrak{g}^\vee)^G$
 which gives the desired isomorphism.

$$\begin{array}{ccc}
 \text{Kos}^\bullet(\mathfrak{g}^\vee) & \xrightarrow{\sim} & \Omega^\bullet(\Omega^1(-, \mathfrak{g})) \\
 \uparrow & & \uparrow \text{from Part 2} \\
 \text{Sym}^\bullet(\mathfrak{g}^\vee)^G & \xrightarrow{\sim} & \Omega^\bullet(\text{Bun}_G^\nabla)
 \end{array}$$

from Part 3

Some homotopical constructions in an arbitrary ∞ -topos $\mathcal{C}$.

1. A monoid object $M : \Delta^{op} \rightarrow \mathcal{C}$ such that

(i) *Segal condition*

$$\forall n_1, n_2 \quad \begin{array}{ccc} [n_1 + n_2] & \xleftarrow{+0} & [n_1] \\ \uparrow +n_1 & & \uparrow +n_1 \\ [n_2] & \xleftarrow{+0} & [0] \end{array}$$

$$\begin{array}{ccc} M_{n_1+n_2} & \longrightarrow & M_{n_1} \\ \downarrow & \lrcorner & \downarrow \\ M_{n_2} & \longrightarrow & M_0 \end{array}$$

is a pullback.

chains of n_1 arrows, chains of n_2 arrows
at a common point

(ii) $M_0 \xrightarrow{\sim} *$ (contractible)

Therefore $M_n \simeq M_1 \times M_1 \times \dots \times M_1$

$$\dots M_1 \times M_1 \begin{array}{c} \xrightarrow{p_1} \\ \xleftarrow{p_2} \end{array} M_1 \xrightarrow{u} *$$

2. A groupoid object $G : \Delta^{op} \rightarrow \mathcal{C}$ such that
 $\forall n_1, n_2$ and all squares i_1, i_2, p_1, p_2

$$\begin{array}{ccc} [n_1 + n_2] & \xleftarrow{i_1} & [n_1] \\ i_2 \uparrow & & \uparrow p_1 \\ [n_2] & \xleftarrow{p_2} & [0] \end{array}$$

- such that
- i_1 and i_2 are injections
 - $\text{Im}(i_1) \cup \text{Im}(i_2) = [n_1 + n_2]$
jointly surjective
 - $\text{Im}(i_1) \cap \text{Im}(i_2) = \{i_1 p_1(0)\}$
singleton

the induced square

$$\begin{array}{ccc} G_{n_1+n_2} & \longrightarrow & G_{n_1} \\ \downarrow & \lrcorner & \downarrow \\ G_{n_2} & \longrightarrow & G_0 \end{array}$$

is a pullback square.

3. A group object in $\mathcal{C}$ is a groupoid object such that $G_0 \xrightarrow{\sim} *$.

4. Given a monoid object M in $\mathcal{C}$, a left M -module in $\mathcal{C}$ is a pair $(A : \Delta^{\text{op}} \rightarrow \mathcal{C}, p : A \rightarrow M)$ such that

$$\begin{array}{ccc} A_n & \xrightarrow{p_n} & M_n \\ \downarrow & \lrcorner & \downarrow \\ A_0 & \xrightarrow{p_0} & M_0 \end{array}$$

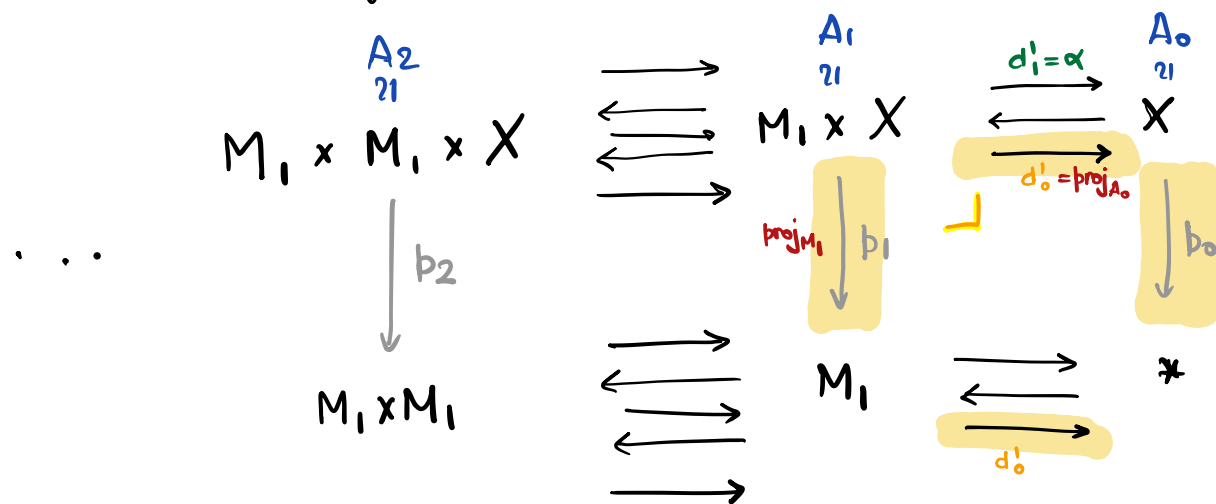
$\begin{array}{c} [n] \\ \uparrow +n \\ [0] \end{array}$

 $\begin{array}{c} [n] \\ \uparrow +n \\ [0] \end{array}$

Note that $M_0 \simeq *$

Therefore $A_n \simeq M_n \times A_0 \simeq M_1 \times \dots \times M_1 \times A_0$

5. Given an object X in $\mathcal{C}$, a left action of M on X is a left M -module A and an equivalence $X \simeq A_0$.



$\alpha : M_1 \times X \longrightarrow X$
is the action.

and higher simplices encode coherences.

6. Trivial action

A monoid M acts on $*$ by taking $A = M$, $\text{id}: M \rightarrow M$

This is the terminal object in $\text{LMod}_M(\mathcal{C})$.

$\text{LMod}_M(\mathcal{C})$ is a full sub-category of $\text{Func}(\Delta^{\text{op}}, \mathcal{C})/M$
on left M -module objects.

$$7. \quad \begin{array}{ccc} | & | & : \text{Func}(\Delta^{\text{op}}, \mathcal{C}) \rightarrow \mathcal{C} \\ & X & \mapsto \text{colim}(\dots X_2 \begin{smallmatrix} \xrightarrow{\quad} \\ \xleftarrow{\quad} \end{smallmatrix} X_1 \begin{smallmatrix} \xrightarrow{\quad} \\ \xleftarrow{\quad} \end{smallmatrix} X_0) \end{array}$$

This is called geometric realisation.

8. Given an object X in $\mathcal{C}$ with a left M -action A ,
define homotopy quotient $X_{//M} := |A|$.

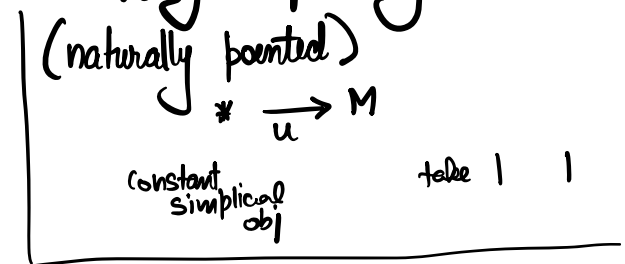
Intuitively $\left\{ \begin{array}{l} \text{colim}(\dots M_! X \begin{smallmatrix} \xrightarrow{\alpha} \\ \xleftarrow{\text{proj}_X} \end{smallmatrix} X) \text{ gives } \begin{array}{l} \text{coequalizer of action and proj} \\ \text{in Sets.} \\ X / \alpha(m, x) \sim \text{proj}(m, x) \\ m \cdot x \sim x \end{array} \end{array} \right\}$

orbits of M -action

Particularly $*_{//M} = |M|$ is called $B_e M$, the classifying object of M in $\mathcal{C}$.

There is a functor

$$-_{//M} : LMod_M(\mathcal{C}) \longrightarrow \mathcal{C}_{/B_e M}$$



9. Given a morphism $e: X \rightarrow Y$ in $\mathcal{C}$

The Čech nerve of e , denoted $\check{C}(e): \Delta^{op} \rightarrow \mathcal{C}$

$$[n] \mapsto X \times_Y \cdots \times_Y X$$

$$\cdots X \times_Y X \times_Y X \begin{matrix} \xrightarrow{\text{proj}_2} \\ \xrightarrow{\text{diag}} \\ \xrightarrow{\text{proj}_1} \end{matrix} X \times_Y X \begin{matrix} \xrightarrow{\text{proj}_2} \\ \xrightarrow{\text{diag}} \\ \xrightarrow{\text{proj}_1} \end{matrix} X$$

Note: $\check{C}(e)$ is a groupoid object.

There is a map $|\check{C}(e)| \longrightarrow Y$

$\left\{ \begin{array}{l} \text{captures the} \\ \text{notion of} \\ \text{covers of } Y \end{array} \right\}$

10. e is an effective epimorphism if $|\check{C}(e)| \simeq Y$

Let $\text{Eff}(\mathcal{C})$ be the full-subcategory of $\text{Func}(\Delta', \mathcal{C})$ on effective epimorphisms.

Thm 1. $\chi : \text{Eff}(\mathcal{C}) \xrightarrow{\sim} \text{Grpd}(\mathcal{C})$ is an equivalence.

backwards map is given by

$$\boxed{G_0 \twoheadrightarrow |G|} \longleftrightarrow G$$

Thm 2. let G be a group object in $\mathcal{C}$

$$-//_G : \text{LMod}_G(\mathcal{C}) \xrightarrow{\sim} \mathcal{C}/_{B_e G}$$

is an equivalence.

backwards map is given by

$$\boxed{\left(\underbrace{\chi(* \rightarrow B_e G)}_{\substack{\uparrow \\ G}} \times X, \text{proj}_G \right)} \longleftrightarrow \boxed{X \xrightarrow{f} B_e G}$$

Think of actions via classifying objects.

Observation

$$\mathcal{C}_{/B_e G} \longrightarrow \text{LMod}_G(\mathcal{C}) \xrightarrow[\text{forget action}]{\text{ev}|_{[0]}} \mathcal{C}$$

$$f: X \rightarrow B_e G \longmapsto \begin{matrix} * & x & X \\ & B_e G & \\ & \text{fib } f & \end{matrix}$$

$$\text{i.e.} \quad \begin{array}{ccc} \text{fib}(f) & \xrightarrow{\quad} & X \\ \downarrow & \lrcorner & \downarrow f \\ * & \longrightarrow & B_e G \end{array}$$

$$X \simeq \text{fib } f // G$$

let Y be an object of $\mathcal{C}$ and $A: \Delta^{\text{op}} \rightarrow \mathcal{C}$ be an action of G on Y . There is a natural equivalence

$$Y \xrightarrow{\simeq} \text{fib}(Y // G \rightarrow B_e G)$$

Ref
Principal ∞ -bundles
Nikolaus, Schreiber, Stevenson (2023)

Thm 3 If $\mathcal{C} = \text{Sh}(\text{Man}, \infty\text{-Grpd})$

let $f : E_1 \rightarrow E_2$. If for $\forall n \geq 0$,

$\pi_0 E_1(\mathbb{R}^n) \xrightarrow{\pi_0(f)} \pi_0 E_2(\mathbb{R}^n)$ is a surjection

then f is an effective epimorphism.

HTT Prop 7.2.1.14 +
more

let $\mathcal{C} = \text{Sh}(\text{Man}, \infty\text{-Grpd})$

$$\text{Man} \hookrightarrow \text{Sh}(\text{Man}, \infty\text{-Grpd})$$

$$M \mapsto \text{Hom}_{\text{Man}}(-, M) = \mathfrak{z}(M) \quad \left[\begin{array}{c} \text{set valued} \\ \text{or} \\ \text{discrete} \end{array} \right]$$

for a Lie group G

$\text{Bar}(G) : \Delta^{\text{op}} \rightarrow \text{Sh}(\text{Man}, \infty\text{-Grpd})$ is a group object.

$$[n] \mapsto \underbrace{\mathfrak{z}(G) \times \dots \times \mathfrak{z}(G)}_{n\text{-fold}}$$

Part 1

Consider $\text{Bun}_G \in \text{Sh}(\text{Man}, \infty\text{-Grpd})$

and $\text{triv}_G : * \longrightarrow \text{Bun}_G$

$\text{triv}_G(M) : * \longrightarrow \text{Bun}_G(M)$

$$* \longmapsto \begin{array}{c} M \times G \\ \downarrow \text{pr}_M \\ M \end{array}$$

for each manifold M
trivial bundle

Prop 1. triv_G is an effective epimorphism

Proof. Using Thm 3, check $\pi_0(\text{triv}_G(\mathbb{R}^n))$ is surjective $\forall n \geq 0$

$$* \longrightarrow \pi_0(\text{Bun}_G(\mathbb{R}^n))$$

?!

every principal G -bundle
on a contractible manifold
is trivializable

Prop 2 (i) $\mathcal{X}(\text{triv}_G) \simeq \text{Bar}(G)$

Proof. $\mathcal{X}(\text{triv}_G)_n =$

$$* \xleftarrow{n+1} * \xrightarrow{\quad} * \xrightarrow{\quad} * \xrightarrow{\quad} * \xrightarrow{\quad} *$$

$\text{Bun}_G \quad \text{Bun}_G \quad \text{Bun}_G$

for $n=1$

$$\begin{array}{ccc} P & \xrightarrow{\quad} & * \\ \downarrow & \lrcorner & \downarrow \\ * & \longrightarrow & \text{Bun}_G \end{array}$$

pointwise

$$\begin{array}{ccc} P(M) & \xrightarrow{\quad} & * \\ \downarrow & \lrcorner & \downarrow \\ * & \longrightarrow & \text{Bun}_G(M) \end{array}$$

$$PCM) = \text{Aut}(M \times G \xrightarrow{\text{pr}_M} M) \quad \text{in } \text{Bun}_G M$$

$$\text{Aut}(M \times G \xrightarrow{\text{pr}} M) \cong \text{Hom}_{\text{Man}}(M, G)$$

$$\begin{array}{ccc} f & \longmapsto & \text{proj}_G \circ f|_{(-, e)} \\ (m, g) & \longleftarrow & \sim \\ \downarrow & & f \\ (m, g \cdot f(m)) & & \end{array}$$

$$P(M) \cong \mathcal{L}(G)(M)$$

$$\therefore P \cong \mathcal{L}(G)$$

$$\text{Similarly } \mathcal{L}(\text{triv}_G)_n \cong \underbrace{\mathcal{L}(G) \times \dots \times \mathcal{L}(G)}_n \cong \text{Bar}(G)_n$$

$$\text{Prop 3 : } \text{Bun}_G \cong {}^* // G$$

Proof : Take geometric realisation on both sides.

$$|\mathcal{L}(\text{triv}_G)| \cong |\text{Bar}(G)| \stackrel{\text{by definition}}{=} {}^* // \text{Bar}(G)$$

But $\mathcal{L}(G)$ is 0-truncated and therefore $\text{Bar}(G)$ is the same data as G .

Also using Thm 1

$$|\mathcal{L}(\text{triv}_G)| \cong \text{Codomain}(\text{triv}_G) = \text{Bun}_G$$

$$\text{Therefore : } \text{Bun}_G \cong {}^* // G \stackrel{\text{by defn.}}{=} B_{\text{Sh}(\text{Man}, \infty\text{-Grp})}^G$$

Consider $\text{Bun}_G^\nabla \in \text{Sh}(\text{Man}, \infty\text{-Grpd})$

$M \mapsto$ principal G -bundle over M with connection
and connection preserving isomorphisms.

there is a map

$$U : \text{Bun}_G^\nabla \rightarrow \text{Bun}_G \quad \text{forget the connection}$$

Then by Thm 2 $\text{Bun}_G^\nabla \simeq \text{fib}(U) // G$

To compute $\text{fib}(U)$

Consider

$$\begin{array}{ccc} \text{Bun}_G^{\nabla, \text{triv}} & \xrightarrow{\quad} & \text{Bun}_G^\nabla \\ \downarrow & \lrcorner & \downarrow U \\ * & \longrightarrow & \text{Bun}_G \end{array}$$

pointwise

$$\begin{array}{ccc} \text{Bun}_G^{\nabla, \text{triv}}(M) & \longrightarrow & \text{Bun}_G^\nabla(M) \\ \downarrow & \lrcorner & \downarrow \\ * & \longrightarrow & \text{Bun}_G(M) \end{array}$$

explicit description
as pullback of groupoids.

$$\text{obj} (P, \oplus, \varphi: M \times G \xrightarrow{\sim} P)$$

$$\text{mor} (P_1, \oplus_1, \varphi_1) \longrightarrow (P_2, \oplus_2, \varphi_2)$$

is $f: P_1 \xrightarrow{\sim} P_2$ an iso of G -bundles

such that $f^*(\oplus_2) = \oplus_1$ and

$$\begin{array}{ccc} & M \times G & \\ \varphi_1 \swarrow & & \searrow \varphi_2 \\ P_1 & \xrightarrow{f} & P_2 \end{array}$$

i.e. $f = \varphi_2^{-1} \circ \varphi_1$

This is in fact 0-truncated (sheaf of sets)

$$\text{define } \text{triv}_G^\nabla : \Omega'(-, \mathfrak{g}) \rightarrow \text{Bun}_G^\nabla$$

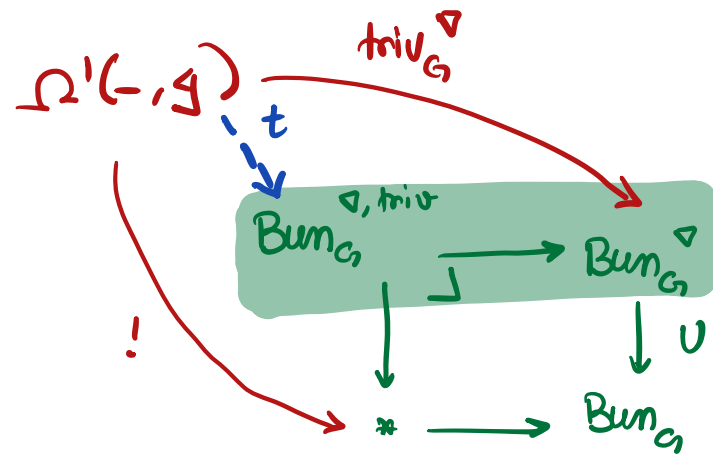
$$\text{triv}_G^\nabla(M) : \Omega'(M, \mathfrak{g}) \rightarrow \text{Bun}_G^\nabla$$

$$\omega \longmapsto \begin{array}{c} M \times G \\ \downarrow \text{pr}_M \\ M \end{array}$$

with connection
determined by ω .

$$\text{pr}_M^*(\omega) + \text{pr}_G^*(\overline{\omega})$$

$$\begin{aligned} M &\subset \Omega'(\mathfrak{g}, \mathfrak{g}) \\ &\mathfrak{g}^{-1} d\mathfrak{g} \\ &(L\mathfrak{g}^{-1})_*(v) \end{aligned}$$



$$\begin{aligned}
 t_M: \Omega^1(M, \mathfrak{g}) &\longrightarrow \text{Bun}_G^{\nabla, \text{triv}}(M) \\
 \omega &\longmapsto (M \times G, \text{pr}_M^*(\omega) + \text{pr}_G^*(MC), \text{id}: M \times G \downarrow M \times G)
 \end{aligned}$$

this is fully faithful because both groupoids are discrete
essentially surjective because of property of connections on trivial bundles.

$\therefore t$ is an equivalence.

We have computed $\text{fib}(U)$.

And $\text{Bun}_G^{\nabla} \simeq \Omega^1(-, \mathfrak{g}) // G$

explicit action

$$\text{Ad}: \mathfrak{g}(G) \times \Omega^1(-, \mathfrak{g}) \rightarrow \Omega^1(-, \mathfrak{g})$$

$$g \cdot \alpha = g^*(MC) + \text{Ad}_{g^{-1}} \alpha$$

Part 2.

Define $\Omega^\bullet : \mathbf{PSh}(\mathbf{Man}, \infty\text{-Grp})^{\text{op}} \rightarrow \mathbf{Vect}_{\mathbb{R}}$

$\mathbb{R} \longmapsto \lim_{\mathbb{Z}(M) \rightarrow F} \Omega^\bullet(M)$

let a lie group G act on a sheaf F .

$$\dots \mathbb{Z}(G) \times \mathbb{Z}(G) \times F \begin{matrix} \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{matrix} \mathbb{Z}(G) \times F \begin{matrix} \xrightarrow{\alpha} \\ \xleftarrow{\text{proj}_F} \end{matrix} F$$

$$\bar{\mathbb{Z}}_g(q, m) = (\bar{\mathbb{Z}}_g(q), 0)$$

$$\mathbb{Z}_g(q) = dR_g|_e(\mathbb{Z}_g)$$

'standard constructions in deRham theory'

F has a G -action, $\mathbb{Z}_g \in \mathfrak{g}$; $\mathbb{Z}_g$ induces an invariant vector field $\mathbb{Z}_g$ on G .

$$\Omega^p(F) \xrightarrow{\alpha_*} \Omega^p(G \times F) \xrightarrow{\mathbb{Z}_g} \Omega^{p-1}(G \times F) \rightarrow \Omega^{p-1}(\{e\} \times F) = \Omega^{p-1}(F)$$

$\mathbb{Z}_g$

'slight horizontal movement'

Thm 4: $\mathcal{F} \longrightarrow \mathcal{F}_{//G}$ induces a map $\Omega^\bullet(\mathcal{F}_{//G}) \rightarrow \Omega^\bullet(\mathcal{F})$

This map is injective at each level and the image is 'basic' subcomplex

$$(i) \quad \forall g \in G \quad \alpha^*(\omega) \big|_{\{g\} \times \mathcal{F}} = \omega \quad \text{"G-invariant"}$$

$$(ii) \quad \forall g \in G \quad \iota_g \omega = 0$$

$$\begin{aligned} \text{Compute } \Omega^p(\mathcal{F}_{//G}) &= \Omega^p(\operatorname{colim} (\dots \overset{\alpha}{\underset{\text{proj}}{\rightleftarrows}} G \times \mathcal{F} \overset{\alpha}{\underset{\text{proj}}{\rightleftarrows}} \mathcal{F})) \\ &\simeq \lim (\dots \Omega^p(\mathcal{F}) \overset{\Omega^p(\alpha)}{\underset{\Omega^p(\text{proj})}{\rightleftarrows}} \Omega^p(G \times \mathcal{F}) \dots) \quad \text{for discrete schemes.} \\ &\simeq \lim (\Omega^p(\mathcal{F}) \rightrightarrows \Omega^p(G \times \mathcal{F})) \quad \text{because these are sets.} \\ &\simeq \ker (\Omega^p(\alpha) - \Omega^p(\text{proj}_{\mathcal{F}})) \hookrightarrow \Omega^p(\mathcal{F}) \end{aligned}$$

Prop. The equalizer is exactly the 'basic' sub-complex.

Part 3.

Defn. Let V be a $\mathbb{R}$ -vector space.

The Koszul complex

$$\text{Kos}^\bullet V \text{ is a dga} = \Lambda^\bullet V \otimes \text{Sym}^\bullet V$$

$$\text{with differential } d_K(v) = \tilde{v}$$

$$d_K(\tilde{v}) = 0$$

$$v \in V = \Lambda^1 V \quad \text{deg 1}$$

$$\tilde{v} \in V = \text{Sym}^1 V \quad \text{deg 2}$$

$\text{Kos}^\bullet V$ is generated by $V = \Lambda^1 V$.

Consider the map

$$\eta : \text{Kos}^\bullet(V^\vee) \longrightarrow \Omega^\bullet(\Omega^1(-, V))$$

$$\text{defined by } \eta^1(l)_m(\sum_i \alpha_i \otimes v_i) = \sum_i l(v_i) \alpha_i$$

$$l \in V^\vee = \Lambda^1(V^\vee)$$

$$\text{for } \alpha_i \in \Omega^1(M) \\ v_i \in V$$

Thm 5. η is an isomorphism of dga.

This is equivalent to saying that every natural transformation

$$\omega : \Omega^1(-, V) \rightarrow \Omega^p(-)$$

is (uniquely) of the form,

$$\omega_M \left(\sum_i \alpha_i \otimes v_i \right) = \sum_{I, J} M_{I, J} (v_{i_1}, \dots, v_{i_k}, v_{j_1}, \dots, v_{j_l}) \alpha_{i_1} \wedge \dots \wedge \alpha_{i_k} \wedge d\alpha_{j_1} \wedge \dots \wedge d\alpha_{j_l}$$

↑
anti-symm in first k variables
and symm in last l .

and $p = k + 2l$

Proof Sketch

1. Uniqueness :

If it is known to be of given form
extract $M_{I, J}$ by evaluating ω for
manifolds of large dimension.
(enough linearly independent 1-forms).

2. lemma - for $\omega \in \Omega^p(\Omega^1(-, V))$ and $\alpha \in \Omega^1(M, V)$
the value of $\omega_N(\alpha)$ at $x \in M$ depends only
on the N -jet of α at x for some N .
Moreover, $N=p$ suffices.

3. ω is determined by its value at the origin of
vector spaces $W \in \text{Man}$.

$$\tilde{\omega}_W : \text{Jet}^p(W; W^V \otimes V) \rightarrow \Lambda^p W^V.$$

$$\bigoplus_{j=0}^p (\text{Sym}^j W^V \otimes W^V \otimes V)$$

lemma - $\tilde{\omega}_W$ is a polynomial of degree $\leq p$.

4. $\tilde{\omega}_W$ gives a linear map

$$T_W : \bigoplus_{i=0}^p \text{Sym}^i \left(\bigoplus_{j=0}^p \text{Sym}^j W^\vee \otimes W^\vee \otimes V \right)$$

$$\downarrow$$

$$\Lambda^p W^\vee$$

this is $GL(W)$ -invariant by naturality in W .

5. (Weyl) - A non-zero $GL(W)$ -invariant linear map $\bigotimes^N W^\vee \rightarrow \Lambda^p W^\vee$ has $N=p$ and is a multiple of the antisymmetrization map.

6. T_w factors through $\bigoplus_{l=0}^p \text{Sym}^l(W^\vee \otimes V \oplus \wedge^2 W^\vee \otimes V)$

therefore $\tilde{\omega}_w$ is polynomial in $\underbrace{W^\vee \otimes V}_{\substack{0^{\text{th}} \text{ jet} \\ \text{value} \\ \alpha}} \oplus \underbrace{\wedge^2 W^\vee \otimes V}_{\substack{1^{\text{st}} \text{ jet} \\ \text{first derivative} \\ d\alpha}}$

Part 4.

We have

$$\eta : \text{Kos}^\bullet(\mathfrak{g}^\vee) \xrightarrow{\simeq} \underbrace{\Omega^\bullet(\Omega'(-, \mathfrak{g}))}_{21} \\ \Omega^\bullet(\text{Bun}_G^{\nabla, \text{triv}})$$

To compute $\Omega^\bullet(\Omega'(-, \mathfrak{g})_{\parallel G}) \simeq \Omega^\bullet(\text{Bun}_G^{\nabla})$

We apply Thm 4.

given $\omega \in \Omega^*(\Omega'(-, \mathfrak{g})) \simeq \text{Kos}^*(\mathfrak{g}^\vee)$

$\omega \in \Omega^*(\text{Bun}_G^\nabla)$ if it satisfies these conditions

$$(i) \quad \forall g \in G \quad \alpha^*(\omega)|_{\{g\} \times F} = \omega$$

$$(ii) \quad \forall \mathfrak{g} \in \mathfrak{g} \quad \iota_{\mathfrak{g}} \omega = 0$$

(i) gives G -invariance condition and restricts us to $(\text{Kos}^*(\mathfrak{g}^\vee))^G$. $A \in \Omega'(M, \mathfrak{g})$

(ii) $\ast \ell \in \wedge^1 \mathfrak{g}^\vee$, then $\eta(\ell)_M(A) = \langle A, \ell \rangle$

$$i_{e_{\mathfrak{g}}} \eta(\ell) = \ell(e_{\mathfrak{g}})$$

$\ast \tilde{\ell} \in \text{Sym}^1 \mathfrak{g}^\vee$ then $\eta(\tilde{\ell})_M(A) = \langle dA, \ell \rangle$ adjoint action on $\mathfrak{g}^\vee$.

$$i_{e_{\mathfrak{g}}} \eta(\tilde{\ell}) = -\text{Ad}_{e_{\mathfrak{g}}}^* \ell \in \wedge^1 \mathfrak{g}^\vee$$

We can compute that $\Omega_L := \tilde{L} + \frac{1}{2} [L, L]$ — curvature

$i_{\xi} \eta(\Omega_L) = 0$ and in fact $\ker i_{\xi} \circ \eta$
is the subcomplex of $\text{Kos}^\bullet(\mathfrak{g}^V)$ generated by

Ω_L .
Since Ω_L is deg 2, this is
 $\text{Sym}^\bullet(\Omega_L \mid L \in \mathfrak{g}^V) \cong \text{Sym}^\bullet(\mathfrak{g}^V)$
base change
 $L \mapsto \Omega_L$

Therefore
CW : $(\text{Sym}^\bullet(\mathfrak{g}^V))^G \xrightarrow{\cong} \Omega_{\text{cl}}^\bullet(\text{Bun}_G^\Delta)$

because $d = 0$

Winkler

given an invariant polynomial of degree k on $\mathfrak{g}$, we apply it to the curvature of a G -connection to obtain a closed $2k$ -form. This assignment is local and natural.